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What is SIS? Difference between SIS and ESD

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An instrumented safety function, or SIF, is one or more components designed to perform a specific task related to safety in the case of a specific hazardous condition.

A Safety Instrumentation System, or SIS, is a collection of SIFs designed to bring an industrial process to a safe condition in case of any dangerous condition detected.

SIS is normally implemented using security PLCs (programmable logic controllers). As its name suggests, it can be programmed so that the logical implementation is much easier compared to electromechanical relays or solid state relays.

SIS must be designed in accordance with strict safety performance criteria to eliminate hidden faults that could lead to failures upon

request. This measure of performance is technically defined by IEC61508 / 61511 as PFD (Probability of failure in demand). Alternatively, it can be translated into Security Availability or RRF (Risk Reduction Factor).

Design consideration of SIS

- Knowing the target SIL for each SIF
- Good understanding of the SIF (Instrumented safety function)
- Choice of logical solver architecture (security PLC): 1oo1D, 1oo2D, 2oo3 etc.
- Choice of sensors and final elements, for example. Transmitter with SIL classification, partial stroke valves tested.
- The application of sensors, logical solver (safety PLC), final elements will strictly follow the guidelines of the Safety Manual that are mentioned in the Safety Certification.
- SIL verification in finalized design
- Spurious travel rates
- Online repair and change flexibility.
- Ease of problem solving and maintenance.
- Frequency test tests and procedures.

Difference between SIS and ESD

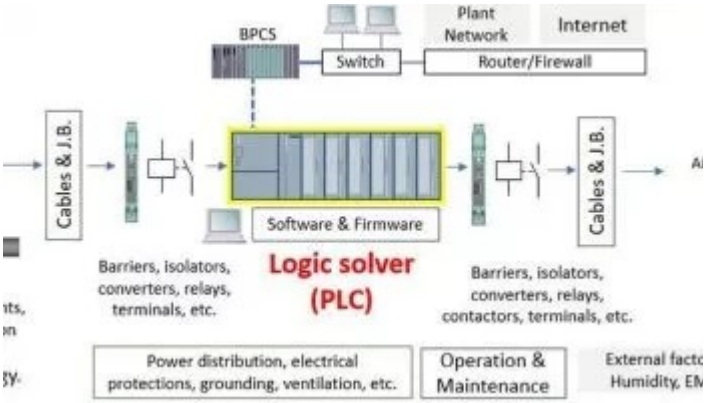
An Emergency Shutdown (ESD) system is a system of manual control stations strategically located on a platform that, when activated, will initiate shutdown of all wells and other process stations.

This system may include a number of independent process shutdown systems that can also be actuated separately. Activation of the ESD system should result in the termination of all production activity on the platform, including the closing of all pipeline SDVs.

The ESD system provides a means for personnel to manually initiate platform shutdown when an abnormal condition is observed. Fusible elements of the fire loop may be integrated with the ESD control loop.

Thus ESD is part of SIS for safety purpose to protect either plant or people and ESD will active when there is someone activated.

Related



[What is SIS (Safety Instrumentation System)?]

Safety instrumented system (SIS) is a system designed to monitor dangerous conditions in a plant (operation unit) and take action in the event of a dangerous condition or a condition where if no action is taken it will cause danger. This system will produce output that will prevent danger or…

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SIS (safety instrumented system) basics

What is SIS? An SIS is safety instrumented system, which takes control action to prevent the risks and safeguard a process when abnormal conditions are present. An SIS loop will be implemented as single or multiple functions, and safety instrumented functions

should be separate from the control hardware used to regulate…

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